



BARKER COLLEGE

TRIAL HIGHER SCHOOL CERTIFICATE 1995

MATHEMATICS

AM TUESDAY 22 AUGUST

3/4 UNIT

TIME: 2 HOURS

[Plus 5 minutes' Reading Time]

DSK

BJR

JCS

BHC

BTP

JM*

125 copies

DIRECTIONS TO CANDIDATES:

1. Write your Candidate Number on EVERY page.
2. Start each question on a NEW page.
3. ALL questions are of equal value.
4. Show ALL necessary working. Marks may be deducted for careless or badly arranged work.
5. Standard integrals are provided at the end of the paper.
6. Board-approved calculators may be used.

* * * *

QUESTION 1.

(a) Solve for x : $\frac{1}{x-1} \leq 3$ [3m]

(b) Find the exact value of $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ [2m]

(c) Find the derivative of $y = \tan^{-1} 2x$ [2m]

(d) Evaluate $\int_0^{\frac{\pi}{6}} \sin^2 2x dx$ [3m]

(e) (i) Sketch the graph of $y = \sin 2x$ for $0 \leq x \leq 2\pi$ [1m]

(ii) Sketch the line $y = \frac{1}{2}$. Without solving the equation $\sin 2x = \frac{1}{2}$, how many solutions are there for the domain $0 \leq x \leq 2\pi$? [1m]

QUESTION 2. [START A NEW PAGE]

(a) $\int_0^1 \frac{x}{1+x} dx$ (using the substitution $u = 1+x$)

(b) Given the function $y = 3 \cos^{-1}\left(\frac{x}{2}\right)$:

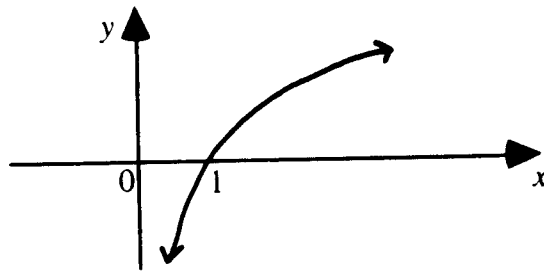
(i) Write down the domain and range.

(ii) Sketch this function.

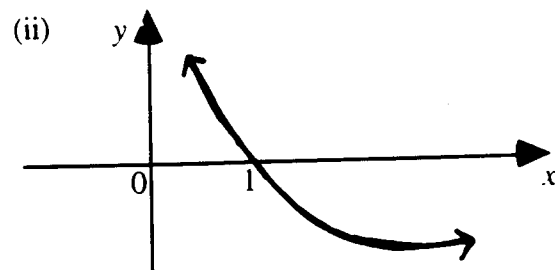
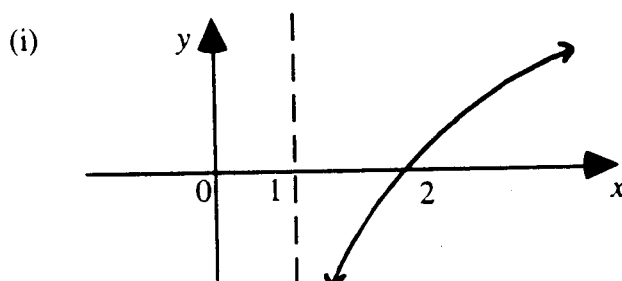
(c) (i) Express $\sin A$ and $\cos A$ in terms of " t " where $t = \tan \frac{A}{2}$

(ii) Hence or otherwise prove that: $\frac{1 + \cos 2A}{\sin 2A} = \cot A$

(d) Given that the following is a sketch of $y = \ln x$



Write down a possible equation for each of the following.



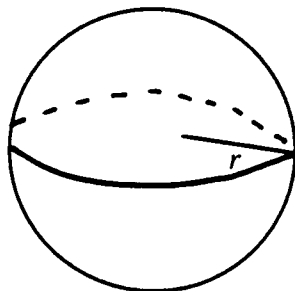
QUESTION 3. [START A NEW PAGE]

- (a) Given the function $y = \frac{2x+1}{x-1}$
- (i) find the domain of this function
 - (ii) what happens to y when $x \rightarrow \infty$
 - (iii) find any vertical and horizontal asymptotes
 - (iv) hence sketch (without calculus) a neat graph of the function. [4m]
- (b) Find the term independent of x in the expansion $\left(2x^3 - \frac{1}{x}\right)^{12}$ [4m]
- (c) Solve the trigonometric equation $2\sin^2 \theta + \sin^2 2\theta = 2$ for $0 \leq \theta \leq 2\pi$ [4m]

QUESTION 4. [START A NEW PAGE]

- (a) Find the acute angle between $2x - y + 5 = 0$ and $y = -3x + 7$ [3m]
- (b) The velocity v m/s of a point moving along the x -axis is given by $v^2 = 16x - 4x^2 + 20$
- (i) Prove that the motion is simple harmonic
 - (ii) Find the centre of motion
 - (iii) Find the length of the path. [4m]

(c)



A spherical balloon is being inflated and its volume increases at a constant rate of 50 mm^3 per second.

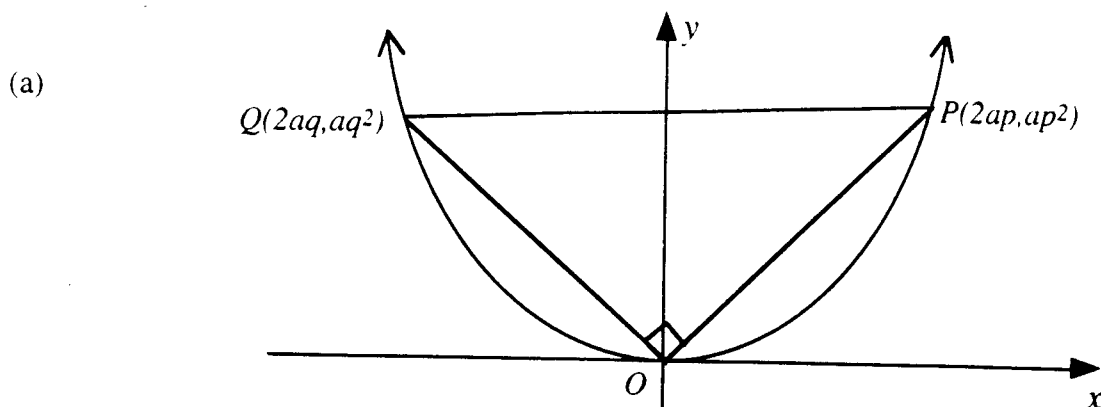
At what rate is its surface area increasing when the radius is 20 mm?

$$V = \frac{4}{3}\pi r^3$$

$$A = 4\pi r^2$$

[5m]

QUESTION 5. [START A NEW PAGE]



PQ is a variable chord of the parabola $x^2 = 4ay$.

It subtends a right angle at the vertex O

If p and q are the parameters corresponding to the points P, Q respectively :

(i) Show that the equation of the tangent to $x^2 = 4ay$ at P is $y - px + ap^2 = 0$

(ii) Hence write down the equation of the tangent at Q , and then find R , the point of intersection of the two tangents drawn from P and Q .

(iii) Find the gradients of PO and QO and hence prove $pq = -4$

(iv) Show that the locus of this point of intersection is $y = -4a$

(b) Use mathematical induction to prove that for all positive integers:

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1)$$

QUESTION 6. [START A NEW PAGE]

(a) A touring cricket side of 15 players contains 5 regular bowlers.

- (i) How many different elevens can be picked which contain exactly 3 of the 5 regular bowlers?
- (ii) What is the probability that if an eleven is picked at random it will only contain 1 regular bowler?
- (iii) What is the probability that if an eleven is picked at random it will contain at least 3 of the regular bowlers? [5m]

(b) (i) Write down the expansion for $(1+x)^n$

(ii) Using this expansion, show that:

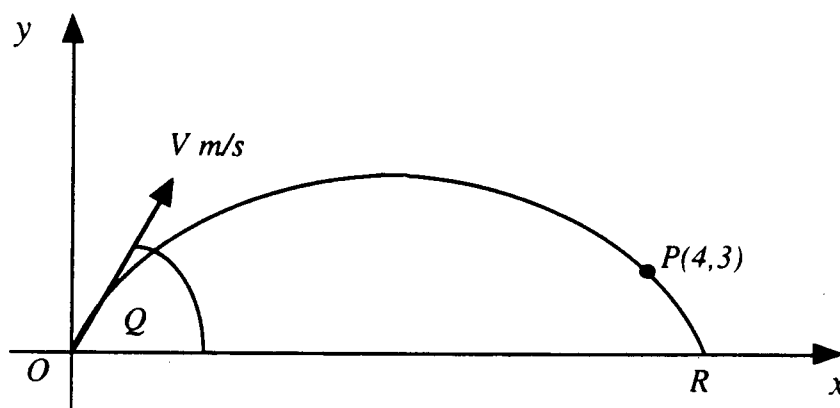
$$\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots = 2^{n-1}$$

(iii) From the identity $(1+x)^n(1+x)^n = (1+x)^{2n}$ compare the coefficient of x^{n+1} on both sides and hence prove that:

$$\binom{n}{0}\binom{n}{1} + \binom{n}{1}\binom{n}{2} + \binom{n}{2}\binom{n}{3} + \dots + \binom{n}{n-1}\binom{n}{n} = \frac{(2n)!}{(n-1)!(n+1)!} \quad [7m]$$

QUESTION 7. [START A NEW PAGE]

- (a) How many 'words' can be formed from AUSTRALIA? (taken all at a time) [2]
- (b) A particle is projected from a point O with a speed of $V \text{ m/s}$ at an angle of θ to the horizontal. Air resistance is to be neglected and $g \text{ m/s}^2$ is the acceleration due to gravity.



- (i) Starting from $\ddot{x} = 0$ show that $x = Vt \cos \theta$ [2]
- (ii) Starting from $\ddot{y} = -g$ show that $y = \frac{-1}{2}gt^2 + Vt \sin \theta$ [2]
- (iii) Prove that the Cartesian equation of path of projectile is given by:

$$y = \frac{-gx^2}{2V^2}(1 + \tan^2 \theta) + x \tan \theta$$

- (iv) You are given that $V^2 = 8g$ and that the particle passes through a point $P(4, 3)$. Hence by using the equation in (iii) find θ , the initial angle of projection and R , the range of the projectile. [2]

END OF PAPER

Q2. a) $\int_0^1 \frac{x}{1+x} dx$

Let $u = 1+x$

$\frac{du}{dx} = 1$

$du = dx$

When $x=1$, $u=2$

When $x=0$, $u=1$

$\int_0^1 \frac{x}{1+x} dx$

$= \int_1^2 \frac{u-1}{u} du$ (1)

$= \int_1^2 (1 - \frac{1}{u}) du$

$= [u - \log u]_1^2$ (1)

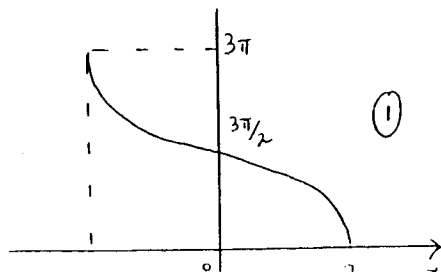
$= 2 - \log 2 - 1 + \log 1$

$= 1 - \log 2.$ (1)

b) i) Domain is $-2 \leq x \leq 2$

Range is $0 \leq y \leq 3\pi$. (1)

ii) $y = 3 \cos^{-1}(\frac{x}{2})$



c) i) If $t = \tan \frac{A}{2}$ then

$\sin A = \frac{2t}{1+t^2}$ (1)

$\cos A = \frac{1-t^2}{1+t^2}$ (1)

ii) Prove $\frac{1 + \cos 2A}{\sin 2A} = \cot A$.

LHS = $\frac{1 + (2\cos^2 A - 1)}{2\sin A \cos A}$

$= \frac{2\cos^2 A}{2\sin A \cos A}$

$= \frac{\cos A}{\sin A} = \cot A = \text{RHS.}$

OR:

LHS = $\frac{1 + \frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}}$

$= \frac{(1+t^2) + (1-t^2)}{2t}$ (1)

$= \frac{1}{t} = \frac{1}{\tan A} = \cot A = \text{RHS.}$ (1)

d) i) $y = \log_e(x-1)$ (1)

ii) $y = -\log_e x$ (1)

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PSK

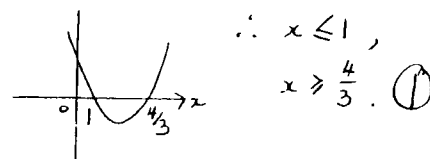
Q1. a) $\frac{1}{x-1} \leq 3$

$x-1 \leq 3(x-1)^2$ (1)

$0 \leq 3(x-1)^2 - (x-1)$

$0 \leq (x-1)[3(x-1)-1]$

$0 \leq (x-1)(3x-4)$ (1)



b) $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$

$= [\sin^{-1} \frac{x}{2}]_1^{\sqrt{3}}$ (1)

$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$

$= \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$ (1)

c) $y = \tan^{-1} 2x$

$y' = \frac{1}{1+4x^2} \cdot 2 = \frac{2}{1+4x^2}$

d) $\int_0^{\pi/6} \sin^2 2x dx$

$= \int_0^{\pi/6} \frac{1}{2}(1 - \cos 4x) dx$ (1)

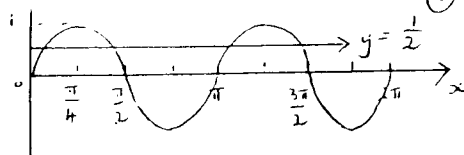
$= \frac{1}{2} [x - \frac{1}{4} \sin 4x]_0^{\pi/6}$

$= \frac{1}{2} [\frac{\pi}{6} - \frac{1}{4} \sin \frac{2\pi}{3}]$ (1)

$= \frac{1}{2} [\frac{\pi}{6} - \frac{1}{4} \cdot \frac{\sqrt{3}}{2}]$

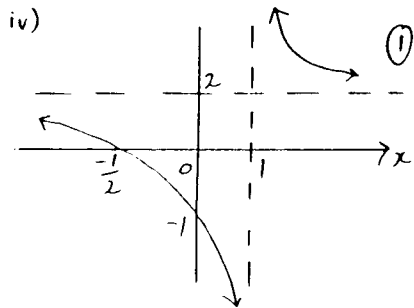
$= \frac{\pi}{12} - \frac{\sqrt{3}}{16} = \frac{4\pi - 3\sqrt{3}}{48}$ (1)

e) i) $y = \sin 2x$



ii) 4 solutions. (1)

- i) Domain is all x , $x \neq 1$. ①
 ii) As $x \rightarrow \infty$, $y \rightarrow 2$. ①
 iii) Vertical asy. at $x = 1$. ①
 Horiz. asy. at $y = 2$. ①



b) The R th term in the expansion of $(2x^3 - \frac{1}{x})^{12}$ is

$$\begin{aligned}
 & {}^{12}C_{R-1} (2x^3)^{13-R} \left(\frac{-1}{x}\right)^{R-1} \quad ① \\
 & = {}^{12}C_{R-1} \cdot (-1)^{R-1} \cdot (2)^{13-R} \cdot x^{39-3R+1-R} \quad ① \\
 & = {}^{12}C_{R-1} \cdot (-1)^{R-1} \cdot (2)^{13-R} \cdot x^{40-4R} \quad ①
 \end{aligned}$$

This is independent of x if $R = 10$
 i.e., $40 - 4R = 0$ is indep. of x . It is

$$\begin{aligned}
 & {}^{12}C_9 \cdot (-1)^9 \cdot 2^3 \\
 & = -1760 \quad ①
 \end{aligned}$$

$$\begin{aligned}
 2(1 - \cos^2 \theta) + \sin^2 2\theta &= 2 \quad ① \\
 2 - 2\cos^2 \theta + \sin^2 2\theta &= 2 \\
 \sin^2 2\theta - 2\cos^2 \theta &= 0 \quad ① \\
 (\sin 2\theta - \sqrt{2} \cos \theta)(\sin 2\theta + \sqrt{2} \cos \theta) &= 0 \\
 \cos^2 \theta (2\sin \theta - \sqrt{2})(2\sin \theta + \sqrt{2}) &= 0 \\
 \cos \theta = 0 \text{ or } \cos \theta = \pm \frac{1}{\sqrt{2}} \quad ① \\
 \theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4} \quad ①
 \end{aligned}$$

has $m_1 = 2$. ①
 $y = -3x + 7$ has $m_2 = -3$.
 Let α be angle between these lines, then
 $\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{2 - (-3)}{1 + 2(-3)} \right| = 1 \quad ①$
 $\therefore \alpha = 45^\circ \quad ①$

b) $v^2 = 16x - 4x^2 + 20$

$$\begin{aligned}
 \text{i) } \frac{1}{2} v^2 &= 8x - 2x^2 + 10 \\
 \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= 8 - 4x \quad ① \\
 &= -4(x - 2) \\
 &= -2^2(x - 2) \quad ①
 \end{aligned}$$

Since $\ddot{x}$ is of the form
 $\ddot{x} = -n^2 X$, $X = x - 2$,
 then motion is SHM.

- ii) Centre is $x = 2$. ①
 iii) length of path is 6 units. ①

$$\begin{aligned}
 A &= 4\pi R^2 \\
 \frac{dA}{dR} &= 8\pi R \\
 \frac{dA}{dt} &= \frac{dA}{dR} \cdot \frac{dR}{dt} \\
 &= 8\pi R \cdot \frac{dR}{dt} \quad ①
 \end{aligned}$$

To find $\frac{dR}{dt}$:

$$V = \frac{4}{3} \pi R^3$$

$$\frac{dV}{dR} = 4\pi R^2$$

$$\frac{dV}{dt} = \frac{dV}{dR} \cdot \frac{dR}{dt} \quad ①$$

$$50 = 4\pi R^2 \cdot \frac{dR}{dt}$$

$$\frac{dR}{dt} = \frac{50}{4\pi R^2} = \frac{25}{2\pi R^2} \quad ①$$

Hence

$$\frac{dA}{dt} = 8\pi R \cdot \frac{25}{2\pi R^2} \quad ①$$

$$= \frac{100}{R}$$

When $R = 20$,

$$\frac{dA}{dt} = 5 \text{ mm}^2/\text{sec.} \quad ①$$

Q5. a)

$$\rightarrow \text{iii) } m \text{ of } OQ = \frac{aq^2 - 0}{2aq - 0}$$

$$= \frac{q}{2}$$

$$m \text{ of } OP = \frac{ap^2 - 0}{2ap - 0}$$

$$= \frac{p}{2}$$

Since $\angle QOP = 90^\circ$ then

$$\frac{q}{2} \times \frac{p}{2} = -1$$

$$\text{ie, } pq = -4 \quad (1)$$

$$\rightarrow \text{i) } y = \frac{1}{4a} x^2$$

$$f'(x) = \frac{x}{2a}$$

$$f'(2ap) = \frac{2ap}{2a} = p$$

Hence eq'n of tangent at P is

$$y - ap^2 = p(x - 2ap) \quad (1)$$

$$y - ap^2 = px - 2ap^2$$

$$\text{iv, } y - px + ap^2 = 0 \quad (1)$$

$$\rightarrow \text{ii) Solve } \begin{cases} y - px + ap^2 = 0 \\ y - qx + aq^2 = 0 \end{cases}$$

$$(1) - (2): -px + qx + ap^2 - aq^2 = 0$$

$$x(q - p) = a(q^2 - p^2) \quad (1)$$

$$\text{ie, } x = a(q + p)$$

$$\therefore y = apq \quad (1)$$

$$R \text{ is } \{a(p+q), apq\}$$

$$\text{iv) } x = a(p+q)$$

$$y = apq$$

$$\text{But } pq = -4, \text{ so}$$

$$y = -4a$$

is locus of R. (1)

b) STEP 1 When $n=1$,

$$\text{LHS} = 1^2 = 1$$

$$\text{RHS} = \frac{1}{6} \cdot 1(2)(3) = 1 \quad (1)$$

Hence the proposition is true when $n=1$.

STEP 2 We assume that

$$1^2 + 2^2 + \dots + k^2 = \frac{1}{6} k(k+1)(2k+1)$$

is true, and now show that

$$1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{1}{6} (k+1)(k+2)(2k+3) \quad (1)$$

$$= \frac{1}{6} (k+1)(k+2)(2k+3)$$

$$\text{Now LHS} = \frac{1}{6} k(k+1)(2k+1) + (k+1)^2$$

$$= \frac{1}{6} (k+1) [k(2k+1) + 6(k+1)]$$

$$= \frac{1}{6} (k+1) [2k^2 + 7k + 6]$$

$$= \frac{1}{6} (k+1)(k+2)(2k+3) \quad (1)$$

$$= \text{RHS.}$$

STEP 3. We have shown that the prop. is true when $n=1$. Assuming that the prop true when $n=k$, we have shown that prop. is true for $n=k+1$ also. Hence the prop. is true when $n=2$, and so true for $n=3$ and so on. Hence it's true for all natural numbers. (1)

Q6. a)

$$\text{i) } {}^5C_3 \times {}^{10}C_8 = 450 \quad (1)$$

ii) With no restrictions, there are

$${}^{15}C_{11} = 1365 \text{ eleven. } (1)$$

The number with only one regular bowler is

$${}^5C_1 \times {}^{10}C_{10} = 5$$

Hence required probability is

$$\frac{5}{1365} = \frac{1}{273} \quad (1)$$

iii) $\textcircled{B}$ Probability

$$= \frac{{}^5C_3 \times {}^{10}C_8 + {}^5C_4 \times {}^{10}C_7 + {}^5C_5 \times {}^{10}C_6}{1365}$$

$$= \frac{1260}{1365} = \frac{12}{13} \quad (1)$$

$$\text{b) i) } (1+x)^n \quad (1)$$

$$= {}^nC_0 x^0 + {}^nC_1 x^1 + \dots + {}^nC_n x^n$$

ii) let $x=1$, then

$$\text{LHS} = 2^n$$

$$\text{RHS} = ({}^nC_0) + ({}^nC_1) + ({}^nC_2) + \dots + ({}^nC_n) \quad (1)$$

Next, let $x=-1$, then

$$\text{LHS} = 0$$

$$\text{RHS} = ({}^nC_0) - ({}^nC_1) + ({}^nC_2) - ({}^nC_3) + \dots$$

$$\text{ie, } ({}^nC_1) + ({}^nC_3) + \dots = ({}^nC_0) + ({}^nC_2) + \dots$$

$$\text{ie, } ({}^nC_1) + ({}^nC_3) + \dots = \frac{1}{2} \times 2^n$$

$$= 2^{n-1} \times 2^n$$

$$= 2^{n-1} \text{ as required. } (1)$$

(iii) Consider $(1+x)^n (1+x)^n = (1+x)^{2n}$

Now the coeff. of x^{n+1} is

$$({}^nC_0)({}^nC_1) + ({}^nC_1)({}^nC_0) + ({}^nC_2)({}^nC_0) + \dots$$

$$+ ({}^nC_n)({}^nC_1)$$

$$= ({}^nC_0)({}^nC_1) + ({}^nC_1)({}^nC_2) + \dots + ({}^nC_{n-1})({}^nC_n)$$

Considering the coeff of x^{n+1}

from $(1+x)^{2n}$ we get

$${}^{2n}C_{n+1} x^{n+1} = \frac{(2n)!}{(n+1)!(2n-(n+1))!} x^{n+1}$$

Hence

$$({}^nC_0)({}^nC_1) + ({}^nC_1)({}^nC_2) + \dots + ({}^nC_{n-1})({}^nC_n) = \frac{(2n)!}{(n+1)!(n-1)!}$$

as required. (1)

$$\frac{3!}{1!} = 60480 \quad \text{①}$$

b) i) $\ddot{x} = 0$

$$\dot{x} = C$$

But when $t = 0$, $\dot{x} = V \cos \theta$

$$\text{so } C = V \cos \theta$$

$$\dot{x} = V \cos \theta$$

$$x = Vt \cos \theta + C_2$$

But $x = 0$ when $t = 0$ so $C_2 = 0$

$$\text{ie, } x = Vt \cos \theta \quad \text{②}$$

ii) $\ddot{y} = -g$

$$\dot{y} = -gt + C_3$$

But when $t = 0$, $\dot{y} = V \sin \theta$

$$\text{so } C_3 = V \sin \theta \text{ so}$$

$$\dot{y} = -gt + V \sin \theta$$

$$y = -\frac{gt^2}{2} + Vt \sin \theta + C_4$$

But $y = 0$ when $t = 0$ so $C_4 = 0$

$$\text{ie, } y = -\frac{gt^2}{2} + Vt \sin \theta \quad \text{②}$$

$$t = \frac{x}{V \cos \theta} \text{ so,}$$

by substitution,

$$y = -\frac{g}{2} \left(\frac{x}{V \cos \theta} \right)^2 + V \left(\frac{x}{V \cos \theta} \right) \sin \theta$$

$$= -\frac{gx^2}{2V^2} \cdot \frac{1}{\cos^2 \theta} + \frac{x \sin \theta}{\cos \theta}$$

$$= -\frac{gx^2}{2V^2} \cdot \sec^2 \theta + x \tan \theta$$

$$= -\frac{gx^2}{2V^2} (1 + \tan^2 \theta) + x \tan \theta$$

as required. ②

iv) From (iii)

$$3 = \frac{-g \cdot 4^2}{2 \cdot 8g} (1 + \tan^2 \theta) + 4 \tan \theta$$

$$3 = -1(1 + \tan^2 \theta) + 4 \tan \theta \quad \text{①}$$

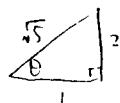
$$\tan^2 \theta - 4 \tan \theta + 4 = 0$$

$$(\tan \theta - 2)(\tan \theta - 2) = 0$$

$$\tan \theta = 2$$

$$\theta = 63^\circ 26'$$

①



To find Range, solve

$$y = 0 \text{ firstly.}$$

$$0 = -\frac{gt^2}{2} + Vt \sin \theta$$

$$\frac{gt^2}{2} = Vt \sin \theta$$

$$t = \frac{2V \sin \theta}{g} \quad \text{①}$$

Sub this into $x = Vt \cos \theta$

and we get

$$R = x = V \left(\frac{2V \sin \theta}{g} \right) \cos \theta$$

$$= \frac{V^2 2 \sin \theta \cos \theta}{g}$$

$$= \frac{V^2 \sin 2\theta}{g} \quad \text{①}$$

$$= \frac{8g \sin 2\theta}{g}$$

$$= 8 \cdot 2 \sin \theta \cos \theta$$

$$= 16 \cdot \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} = \frac{32}{5} \text{ or } 6.4 \text{ m}$$